

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

SP6 - Financial Derivatives

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) You are an investment actuary in XYZ Ltd. The company wants to buy a European Call option on a USD denominated stock without taking a currency risk. A bank has agreed to offer the desired product and given the terms. You have the following data:

Spot price of foreign stock (S)	\$120
Strike price (K)	INR 9000
Time to maturity (T)	1 year
Volatility of foreign stock (σ_s)	25%
Volatility of USD/INR exchange rate (σ_x)	8%
Correlation between the stock and FX rate (ρ)	+0.3
Foreign risk-free rate (r_f)	4%
Domestic risk-free rate (r_d)	7%
USD/INR fixed exchange rate for the deal (X_0)	90

- i) What is the expression for payoff from this option on expiry? (3)
- ii) Derive the adjusted drift rate (r_{adj}) to be used in the Black Scholes model to calculate the price of this option, (7)
- iii) Estimate the theoretical price of the call option using the drift adjusted Black Scholes model derived in (ii) with the necessary adjustments. (5)

[15 Marks]

Q.2) Let S_0 be the initial price of a share of stock that pays no dividends. Let r be equal to {one plus the risk-free rate}, let μ be equal to {one plus the mean stock return}, and let σ be the standard deviation of the stock return, all per period. What is the price today of a claim maturing 10 periods from now that pays the average stock price over the next 10 periods?

[7 Marks]

Q.3)

- i) Describe briefly the Vasicek one-factor model of interest rates and its key statistical properties. (3)
- ii) Let r_t denote the short rate of interest applicable at time t . According to a particular parameterization of this model, the instantaneous forward rate applicable at time $t > 0$ implied by the market prices at time 0 is found to be:

$$f(t) = r_0 e^{-\alpha t} + r_\infty (1 - e^{-\alpha t}) + k(1 - e^{-\alpha t})e^{-\alpha t}$$

where α is a positive constant.

a. Show that, if the graph of $f(t)$ has a stationary point, then

$$e^{-\alpha t} = \frac{r_0 - r_\infty + k}{2k}$$

b. Hence describe the shape of the graph of $f(t)$ in each of the following cases

i. $r_0 = 0.04, r_\infty = 0.06, k = 0.10$

ii. $r_0 = 0.06, r_\infty = 0.04, k = 0.02$

- c. State a possible explanation for a yield curve with the shape of the graph in (ii)(b)(i).

(6)

- iii) Describe briefly the main advantages and limitations of the Vasicek model. (2)

[11 Marks]

Q.4) A trader observed the following prices for the 6 different interest rate derivative contracts mentioned below. The prices are: Rs. 23,000; Rs. 0; Rs. 22,000; Rs. 40,000; Rs. 10,000 and Rs. 28,000 and the contracts are.

- A 4% cap on quarterly interest payments
- A 4% cap on monthly interest payments
- A 4% floor on quarterly interest payments
- A 5% floor on quarterly interest payments
- A 2.5% / 4% collar on quarterly interest payments
- A fixed – for – floating 4% swap on quarterly interest payments

All contracts have a 5-year term and the principal and all other details for each of the 6 alternate contracts mentioned above are the same. Also, the quoted interest rates are compounded with the same frequency as the payments and the current continuously compounded short rate is 3%.

The trader knows that the prices are correct, but they are not necessarily reported in the same order as the contracts are mentioned. The trader has now reached out to you for help. Given this information, match the prices with the relevant contract with proper rationale. State your assumptions clearly.

[10 Marks]

Q.5) An insurance regulator of a country has come up with a new Solvency regulation which leads to a significantly increased contribution of the longevity risk towards capital requirement for a life insurance company. As the investment manager of a life insurance company, you are required to provide a report to the board as to how such risks can be managed.

- i) Describe how would you propose to manage this risk using derivatives – what are the main type of derivative instruments that you propose to use to reduce the longevity risk.

The company had decided to use a longevity swap, though the swap is not the traditional zero start swap. This is due to the fact that company is required to hedge the risk as per regulatory guidelines, which has 10% margin for adverse deviation.

(4)

- ii) What will be the initial value of the swap with a 5-year duration, payment in arrears, of Rs. 100,000 at best estimate assumptions. Assume the risk-free interest rate is 5% with continuous compounding. (6)

- iii) Estimate the reduction in the capital requirement due to longevity swap. The capital requirement for the longevity risk is the difference between base regulatory reserves and reserves assuming the regulatory cashflows increased by 50%. (5)

The regulator has rejected the proposal of the longevity swap expressing concerns with the exposure of counterparty risk.

- iv) Discuss the concerns of the regulators (assuming the transaction is over the counter). State different ways by which your company can mitigate these concerns. (4)

One of the solutions recommended by the regulator is to have sufficient collateral to reduce the counterparty risk.

- v) Estimate the collateral the counterparty A should post such that counterparty risk is minimum. Assume that collateral can cover a 1-in 200 annuity payout scenario where in 1-in-200 scenario cashflow is 50% higher than the base regulatory basis cashflow. You also know that the credit rating of counterparty is A with default assumption, as per an independent credit rating agency, is as follows:

Year	Default Rate
1	2%
2	4%
3	6%
4	8%
5	10%

(6)

[25 Marks]

Q.6)

- i) Connect the Vasicek model with the AR(1) process. (2)
- ii) Suppose that the spot rate curve $r(r, a, b, t, T) = r + a(T - t) + b(T - t)^2$ is implied by a three factor model. Which of the factors, r , a and b affects slope, curvature and parallel moves respectively? (3)
- iii) Between a normal model and log-normal model, which overprices out – of – the money calls on bonds and which underprices out – of – the – money puts on bonds? Why? (2)
- iv) When pricing derivative on bonds under the HJM model, does the tree have to be built over the life of the longer – term underlying bond or just over the life of the derivative. Explain your answer. (2)

[9 Marks]

Q.7) You are an investment actuary offering consulting services specializing in derivatives. Suppose it is May 1, 2025 today and a client comes to you for advice. He is holding 1000 shares of Happy Bank of Utopia (HBU) and he has observed that the futures are trading at a substantial discount to the market. The client wants to sell the stock and buy the futures.

Before you offer your advice or proceed further, you do some research on HBU prices and observe the following (as of May 1, 2025):

- ACU spot price: \$ 650 / \$ 652

- ACU 1 month futures (i.e. expiring on May 31, 2025): \$ 590 / \$ 594
- Interest rate for all terms: 5% / 6%
- Dividend per share: \$ 45
- Ex – Dividend Date: May 15, 2025
- Brokerage on Spot Transaction: 1% of the spot value
- Brokerage on Futures Transaction: 0.5% of the future price
- Margin required at any point in time: 5% of the underlying value as at the close of the day (assume \$ 651)

Further, for the purpose of this question, assume that futures will NOT be settled daily and margin will be released at the maturity along with the settlement.

Based on the above information,

- i) What will be your advice with respect to the soundness of the client's strategy of selling the stock and buying futures? (7)
- ii) Also, what are the risks that the client will be exposed to while implementing the strategy? (2)

[9 Marks]

Q.8) XYZ Ltd. is considering selling a 4-year credit default swap. The initial estimate of default for the company (who issued the bond underlying the CDS) is 4% for the first year but increases by 0.5% each year thereafter. You also assume a recovery rate of 40%, and that any default will take place at the mid-point of the year. The risk-free interest rate is 12%, continuously compounded.

- i) In order to maintain the customer relationship, the Sales department has assured the buyer that the pricing of CDS would assume no default in between year 2 – 3. Estimate the theoretical price of the CDS using the information given. (*Assume premium paid annually in arrear, default take place in mid of year*). (8)
- ii) How much loss would the company be making due to this deal (in spread terms)? (6)

[14 Marks]
